CPSC 471 - Proposal

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Main Idea

Our team will design and implement a transaction-based web application that stores and retrieves its data from a database schema. We will use Python and SQLAlchemy to connect to our chosen database to create, manipulate, and retrieve data from our database schema. Flask will be used as our web framework.

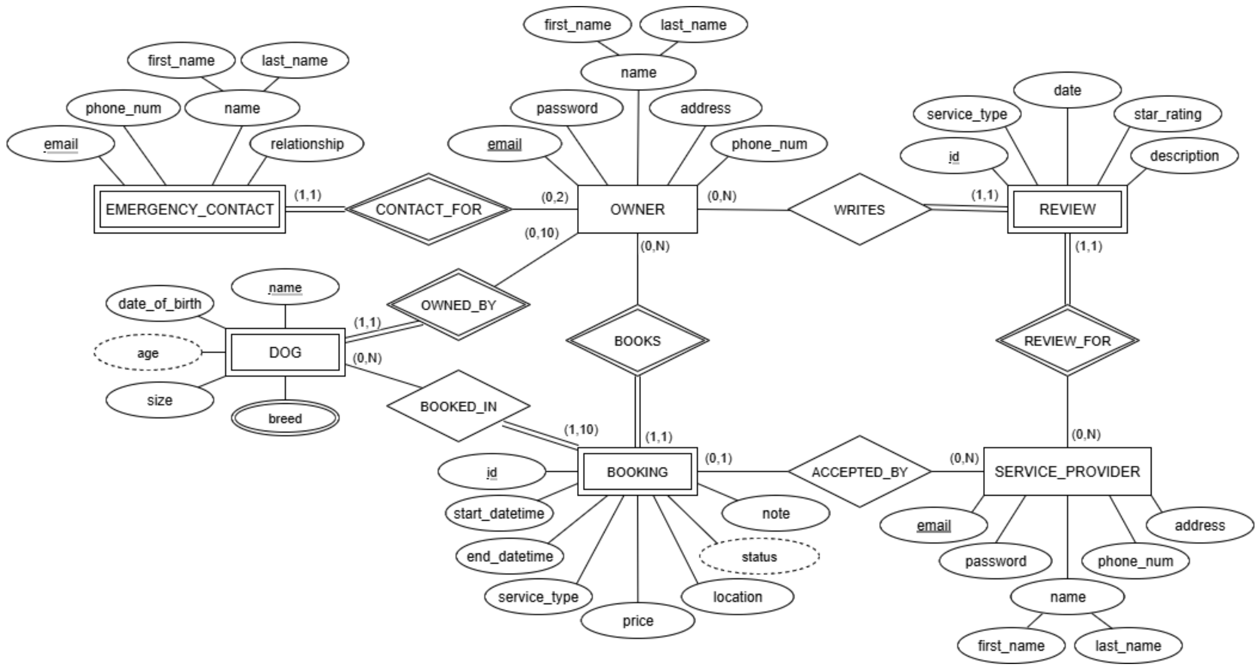
Our web application is called BarkBase. It is a platform for planning and scheduling bookings for dog walks and dog sitting services. Our database primarily includes dog owners, their dogs, emergency contacts, bookings, service providers, and reviews.

Owners are our primary users. They create bookings for their dogs, which are then accepted by service providers. Owners can have emergency contacts, which are used by the service providers during a booking in the event of an emergency where the owner is unable to be contacted. Service providers use our application to accept the booking created by dog owners. They have the freedom to choose as many bookings of whichever types they desire.

BarkBase offers two types of bookings: dog walking and dog sitting. Owners can create dog walking bookings with a meeting location of their choice and with their desired duration. Owners can additionally create dog sitting bookings for their dogs to be cared for over a timespan of their choosing. Upon the completion of a booking, owners are presented with an opportunity to write a review for the service provider of their booking.

From short-term walks to long-term dog sitting, BarkBase covers a variety of needs. We are proud to provide a clean and flexible application for connecting owners with service providers. Our built-in review system provides owners with the confidence they deserve that their dogs are receiving the best care possible. While our application will initially be targeted at owners and service providers within Calgary, we are excited for the future potential to expand our services across the nation.

BarkBase ER Diagram



The following notes and assumptions apply to the above BarkBase ER Diagram:

1. Each owner can have between 0 and 2 emergency contacts. Emergency contacts are used by service providers in the event of an emergency where the owner is unable to be contacted.
2. In the composite name attribute for the emergency contacts, the first name is required and the last name is optional.
3. Owners and service providers have their own BarkBase user accounts, which are logged into using their email and password. Password attributes are stored securely, rather than in plaintext.
4. For simplicity, owners and service providers are kept separate. If a user wishes to take on both roles, they will need to create two separate accounts. This design may be changed in the future to merge the owner and service provider functionality into one general account.
5. Each owner can have between 0 and 10 dogs. The minimum of 0 is to account for new users, and it is assumed that at most 10 dogs need to be supported for each owner.
6. Each dog can only have one owner. For real-world cases where a dog has multiple owners, the owners will either need to share or create separate accounts.

7. The size of a dog is specified as either small, medium, or large in the size attribute on the DOG entity type.
8. Each booking has a price. The price is determined by BarkBase based on the duration and type of service for the booking.
9. Each booking can either be for a dog walk or for dog sitting. The type of service is stored in the service_type attribute on the BOOKING entity type.
10. The location of a booking indicates where the owner and service provider will meet. This may be the owner’s address, the service provider’s address, or a custom location such as the address for a park.
11. The note on a booking is an optional opportunity for owners to provide extra information about their booking to service providers. A booking note could include information such as important details about a dog’s behaviour, or any special requests for the booking.
12. Each booking has a status of either pending, accepted, in progress, or completed. The status is derived from whether a service provider has accepted the booking and whether the booking has ended.
13. After the completion of a booking, owners have the opportunity to write reviews for their service providers. Each review includes the type of service, the date, a star rating out of 5, and an optional description to include in the review.

References

The following references were selected because they combine **technical documentation** with **real-world product examples** that inform the design, implementation, and feature scope of BarkBase. Ordered by relevance, below are our primary references for developing BarkBase.

Flask. (n.d.). Flask documentation. Pallets Projects. Retrieved September 29, 2025, from https://flask.palletsprojects.com/

Flask is the lightweight Python framework we will use to implement our backend services, such as API endpoints for bookings and user management. The official documentation is authoritative and comprehensive, providing examples, tutorials, and detailed explanations of Flask's features. Because our backend will be built with Flask as its web framework, Flask's documentation is crucial for ensuring we apply the framework correctly.

SQLAlchemy. (n.d.). SQLAlchemy documentation. SQLAlchemy. Retrieved September 29, 2025, from https://docs.sqlalchemy.org/

SQLAlchemy is the Object Relational Mapper (ORM) we will use to interact with our PostgreSQL database. It abstracts raw SQL into Python objects, which simplifies database queries and schema management. The official documentation is the primary reference for using SQLAlchemy's API correctly and efficiently, making it an indispensable source for our database implementation.

PostgreSQL Global Development Group. (n.d.). PostgreSQL documentation. PostgreSQL. Retrieved September 29, 2025, from https://www.postgresql.org/docs/

PostgreSQL is the relational database management system we will use to store BarkBase data (users, bookings, dogs, service providers, etc.). The official documentation is the most reliable reference for configuring PostgreSQL locally, setting up connections, and learning about advanced SQL features such as constraints, indexing, and joins. Since database performance and correctness are critical to our project, this is an essential resource.

Rover. (n.d.). Rover: Dog boarding and walking app. Rover.com. Retrieved September 29, 2025, from https://www.rover.com/

Rover is an industry-leading dog walking and sitting platform with features similar to what we plan for BarkBase. Reviewing Rover allows us to study how they structure services, bookings, and user experiences, which can inspire our own design decisions. While it is not a technical source, it provides practical insights into real-world feature sets and user expectations, which makes it valuable for guiding our feature scope.

Wag!. (n.d.). Wag! Dog walking service. WagWalking.com. Retrieved September 29, 2025, from https://wagwalking.com/

Wag! is another popular dog walking and pet care platform. Like Rover, it helps us benchmark our planned application against established competitors. We can analyze their booking flows, service offerings, and interface design to identify both best practices and opportunities for differentiation. This makes Wag! a strong comparative reference for shaping BarkBase's functionality.